Learning a Decoder-Free Depth Encoder for Fixed-Focus Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Optical encoders learned jointly with a reconstruction network are optimised for that network, motivating designs whose gains extend across reconstruction methods. We demonstrate that optimising depth discrimination yields an optical encoder that improves recovery across the tested reconstruction methods. An angle-scanned plane wave passes through a fixed, defocused phase plate, generating depth-dependent excitation patterns while the specimen and detection focus remain stationary. The plate is trained through the optical forward model to minimise weighted squared correlations between detector-integrated depth signatures, without specimen examples or a reconstruction solver. Evaluation uses total variation regularisation and plug-and-play reconstruction with either pretrained or neuron-trained denoisers. At eight-fold measurement compression and equal total excitation dose, the learned plate improves reconstruction of simulated mouse pyramidal neuron volumes relative to its random initialisation. Mean local depth error decreases from 0.26 *to* $0.15\,\mu\mathrm{m}$ *with total variation, from* 0.49 *to* $0.22\,\mu\mathrm{m}$ *with pretrained denoisers, and from* 0.20 *to* $0.14\,\mu\mathrm{m}$ *with neuron-trained denoisers. The gains persist with background and read noise and at reduced dose. The proposed objective also yields* 12–$22\,\%$ *lower depth error than plates trained with worst-case coherence and Fisher-information objectives under the same budget. Together, these results demonstrate improved depth recovery across the tested reconstruction methods without coupling optical training to a decoder.*

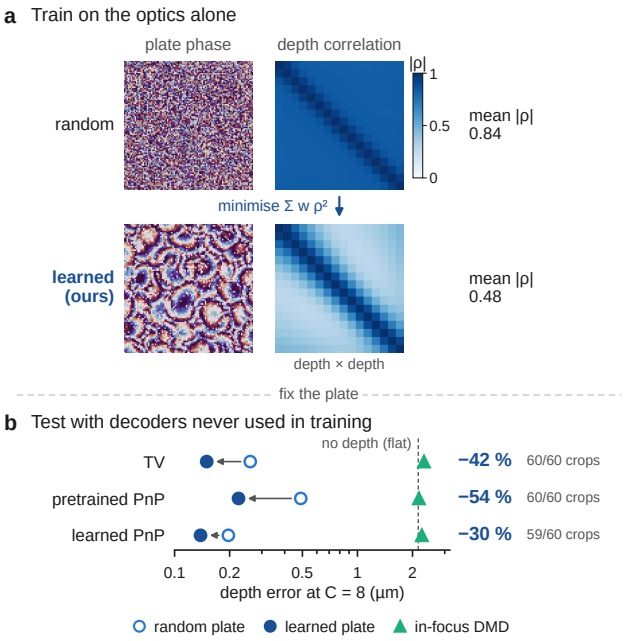


Figure 1. **Decoder-free optical learning improves depth recovery across reconstruction methods.** (a) Phase maps and depth-signature correlation matrices before and after training. (b) The learned plate reduces mean local depth error relative to its random initialisation across TV and two PnP decoders, at eight-fold measurement compression and equal excitation dose. Percentages and counts indicate error reductions and crops improved. The in-focus DMD remains near the flat-depth baseline.

1. Introduction

Computational imaging enables three-dimensional reconstruction from two-dimensional measurements, but recovery depends on how well those measurements distinguish structures at different depths. When different axial arrangements produce similar observations, reconstruction relies more heavily on prior assumptions [3, 7]. In fluorescence microscopy, neuronal branches that overlap in projection may lie at different depths, making this distinction important for recovering their spatial organisation. We investigate how learning the illumination can improve depth discrimination at the measurement stage.

We address this problem by encoding depth in the excitation illumination. A plane wave is scanned in angle through a fixed, defocused phase plate, producing structured illumination patterns that vary with depth (Fig. 2). As the incident angle changes, depth-dependent lateral shifts cause structures at different axial positions to experience different sequences of excitation intensities. A fixed-focus camera records the resulting fluorescence images for computational volume recovery, without axial specimen scanning or detection refocusing.

The effectiveness of this encoding depends on the phase plate. Recent approaches learn optical encoders jointly with reconstruction networks [9, 17, 23], optimising the measurements for a particular decoder. This couples the optical design to a reconstruction method that may change after fabrication. Building on decoder-free optical design [19, 22, 26], we instead optimise the plate through the optical forward model to improve depth discrimination. We minimise weighted squared correlations between detector-integrated measurement signatures of voxels at different depths but the same lateral position. Training requires neither specimen examples nor a reconstruction solver, and we evaluate the resulting encoder across different reconstruction methods.

We evaluate the learned illumination using simulated measurements of mouse pyramidal neuron volumes. Reconstruction uses total variation regularisation and plug-and-play methods with pretrained and neuron-trained denoisers. Across all three methods, the learned phase plate improves mean depth accuracy relative to its random initialisation at the same measurement budget and excitation dose (Fig. 1). We also examine robustness to background and read noise and reduced excitation dose, and compare the training objective with alternative decoder-independent criteria. Our contributions are threefold:

1. **Fixed-focus depth encoding.** We develop an angle-scanned illumination system with a fixed, defocused phase plate and a voxel-integrated vectorial forward model for three-dimensional fluorescence microscopy.
2. **Decoder-free optical optimisation.** We learn the phase plate by minimising weighted squared correlations between detector-integrated depth signatures, without specimen examples or a reconstruction solver.
3. **Evaluation across reconstruction methods.** We demonstrate improved depth recovery with total variation and plug-and-play reconstruction, compare against alternative decoder-free objectives, and assess robustness to noise and reduced excitation dose.

2. Related work

Depth encoding in fluorescence microscopy. Three-dimensional fluorescence microscopy can encode depth through detection or excitation. Light-field microscopy, DiffuserCam and Miniscope3D recover volumes from depth-dependent detection responses [2, 5, 26]. Excitation-based methods instead use illumination patterns that vary with depth. Yeh *et al*. [27] combine translated speckle patterns with axial scanning, while Pascucci *et al*. [18] exploit axial speckle decorrelation for volumetric recovery from a two-dimensional scan. Related approaches use angular scanning through a fixed scattering layer for two-dimensional fluorescence recovery [28], or tilted pattern projection to produce depth-dependent lateral shifts [10]. We combine angular scanning with a fixed, defocused phase plate and learn its phase profile to improve depth discrimination.

Learning optical encoders. Optical design can be guided by reconstruction performance or by properties of the measurements. End-to-end methods jointly optimise optics and reconstruction networks [9, 17, 23], while Kellman *et al*. [12] optimise illumination for a specified reconstruction algorithm. Decoder-independent methods instead optimise optical responses directly. Shechtman *et al*. [22] use Fisher information for 3D localisation, while Miniscope3D [26] optimises a detection-side phase mask by reducing cross-coherence between depth-dependent point-spread functions. Similar coherence objectives appear in sensing-matrix design [6]. IDEAL and IDEAL-IO [15, 19] optimise estimated mutual information using fitted measurement-distribution models, without a reconstruction decoder. Our excitation-side objective minimises weighted squared correlations between detector-integrated depth signatures at each lateral position, without specimen examples or a reconstruction solver.

Volumetric reconstruction and priors. Recovering three-dimensional structure from limited measurements requires assumptions about the specimen. Total variation promotes piecewise-smooth structure [21], while plug-and-play (PnP) methods incorporate denoisers as implicit priors [11, 24]. Multi-slice fusion combines lower-dimensional denoisers acting along complementary orientations of a shared volume [13]. We adapt this framework to fluorescence reconstruction, combining measurement consistency, slice-wise denoising and sparsity regularisation. Stronger priors can improve recovery, but may favour plausible structures that the measurements cannot distinguish [3, 7]. We therefore evaluate optical gains under both conventional regularisation and learned priors, separating improvements in depth encoding from those introduced by reconstruction. Conformal risk control provides a complementary assessment through calibrated depth intervals [1].

3. Angle-scanned phase-encoded 3D imaging

Acquisition. We vary the illumination direction across two angular coordinates while keeping the phase plate fixed (Fig. 2). For transverse position $\mathbf{r} = (x, y)$, the incident plane wave is $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^\top \mathbf{r}}$, with $\mathbf{s}_t = \cos\theta_t(\cos\alpha_t, \sin\alpha_t)$, where α_t is the azimuth and θ_t parametrises the direction in the vertical plane containing that azimuth. The corresponding forward direction is $(\mathbf{s}_t, \sin\theta_t)$, and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ for the simulated refractive index of one. Thus, the two coordinates control the orientation and signed magnitude of the transverse phase gradient. The scan contains 256 distinct directions with $0.5 \leq \|\mathbf{s}_t\| \leq$

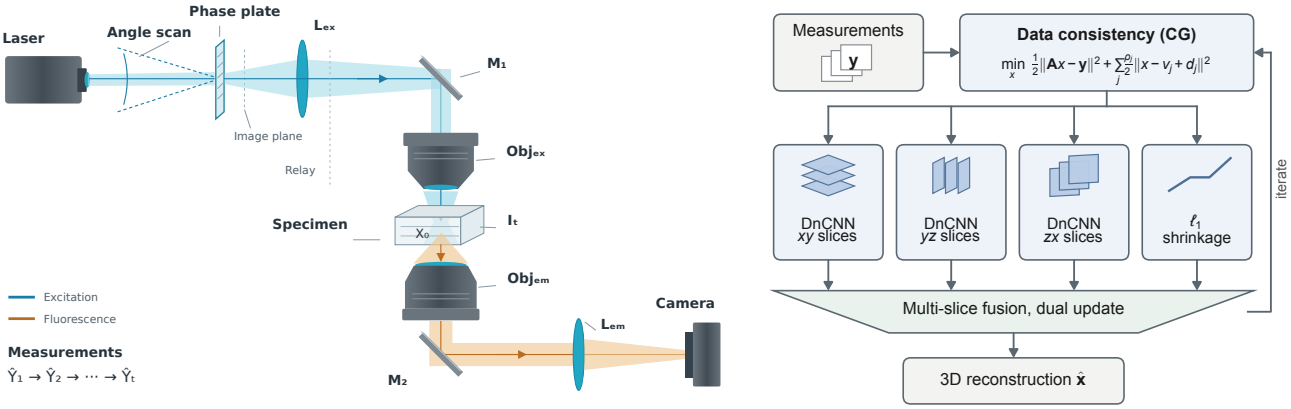


Figure 2. **Fixed-focus acquisition and reconstruction.** Left: a tilted plane wave illuminates a fixed phase plate placed 6.6 μm (specimen-space units) before an intermediate illumination image plane; an NA-0.8 vectorial relay maps the field into the specimen; a camera at a fixed detection focus records one image per illumination angle. Only the input angle changes: plate, specimen and focus are stationary, and nothing is scanned axially. Blue: excitation; orange: fluorescence. Right: one of the three decoders of Sec. 4, multi-slice plug-and-play recovery, alternates a conjugate-gradient data-consistency step with a 2D denoiser applied to the xy, yz and zx slices of a shared volume and an ℓ_1 shrinkage, fused by ADMM.

$\sqrt{3}/2$. Each incident field passes through the phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagates a distance D to the intermediate illumination image plane by angular-spectrum propagation. The NA-0.8 objective pupil filters the modulated field before a vectorial Richards–Wolf relay [14, 20] produces the specimen excitation $I_t = \sum_{c\in\{x,y,z\}} |E_{t,c}|^2$, assuming x-polarised illumination. Coordinates and distances are expressed in specimen-equivalent units. The camera records one fluorescence image per direction at a fixed detection focus. Acquisition is sequential, not single-shot.

Why the angle encodes depth. Propagation after the phase plate produces a spatially structured excitation pattern. Changing the illumination angle shifts this pattern laterally, with displacement increasing with propagation distance. In the paraxial thin-screen approximation, a change $\Delta\mathbf{s}$ in transverse direction cosine produces a displacement $\Delta\mathbf{r} \approx d\,\Delta\mathbf{s}$ after propagation over distance d. Different depths therefore experience different sequences of excitation intensities during the angular scan. Plate defocus places the shear pivot outside the specimen. This model explains the encoding principle, while our high-NA vectorial model accounts for finite-pupil effects and departures from ideal translation. Section 6.1 evaluates the resulting shear.

Image formation. Let $X_0(\mathbf{r}, z)$ denote the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$ [27]. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z')X_0(\mathbf{r}', z')I_t(\mathbf{r}', z')\,\mathrm{d}^2\mathbf{r}'\,\mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity point-spread function and Ω is the specimen volume. Pixel p, occupying region D_p, records the normalised sample $\hat{Y}^0_{t,p} = |D_p|^{-1}\int_{D_p} g_t(\mathbf{r})\,\mathrm{d}^2\mathbf{r}$. Fluorescence contributions therefore add incoherently before pixel averaging. Multiplication by the excitation pattern mixes specimen spatial frequencies before detection blur, so the camera need not resolve the illumination pattern itself. Depth discrimination depends on how these measurements vary across illumination angles.

Voxel-integrated operator. To capture excitation variations within reconstruction voxels, we evaluate the optics on a finer grid and integrate the measurement responses over each voxel. Let B be the fine-grid measurement operator, including emission propagation and pixel averaging, and let U replicate each reconstruction coefficient over its fine-grid cells. Then $\mathsf{A} = \mathsf{B}\mathsf{U}$ preserves the fine-grid response for every specimen represented in this piecewise-constant basis.

Sub-voxel specimen structure remains unrepresented. For a fine-grid specimen $\mathbf{x}_f$, its coarse representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, obtained by voxel averaging, satisfies

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{U}\mathsf{R}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{2}$$

Thus, integration retains within-voxel optical variation, while $\boldsymbol{\delta}_{\mathrm{disc}}$ captures specimen approximation error.

Linear, incoherent fluorescence enables efficient assembly:

$$\mathsf{A}_{(t,p),j} = \sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{3}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells, $(\mathbf{r}_i, z_i)$ is the location of cell i, and $g_{p,i}$ is its emission footprint at pixel p, including its integration weight. These footprints are computed once and reused across angles. After precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF\log F)$ for constructing columns through separate FFT-based forward simulations, with T patterns, P detector samples, F fine-grid cells and N reconstruction voxels.

Learning the plate. We optimise the plate phase ϕ to reduce correlations between measurement signatures at different depths, with the aim of improving depth discrimination. For a voxel at lateral position $\mathbf{r}$ and depth z, let $\mathbf{a}_{\mathbf{r},z}$ denote its centred, unit-norm operator column (Eq. (3)), collecting detector responses across illumination angles. We minimise

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r}} \sum_{z\neq z'} w_{zz'} \left(\mathbf{a}_{\mathbf{r},z}^{\top}\mathbf{a}_{\mathbf{r},z'}\right)^2, \tag{4}$$

where pairs of planes mirrored about the detection focus, $(z, 14-z)$ on the 16-plane grid, receive weight two because they are hardest to separate, and other pairs receive weight one. Squaring penalises correlations of either sign; normalisation prevents uniform amplification of the signatures from reducing the loss.

Training differentiates through the excitation, vectorial relay and detection model, without specimen examples or a reconstruction solver. The objective is a surrogate for depth discrimination, not a guarantee of stable recovery; whether it helps is tested with the decoders of Sec. 4, none of which is used in training. Training details are given in the supplement.

4. Decoders and depth intervals

We evaluate the learned optical encoder using three reconstruction methods: total variation (TV), plug-and-play (PnP) with a pretrained denoiser, and PnP with neuron-trained denoisers. None is trained jointly with the optical encoder. Each method uses the corresponding forward operator, while its prior remains fixed across encoders. Scalar reconstruction parameters are selected separately for each encoder and compression setting using held-out validation volumes.

Total variation. As a non-learned baseline, we reconstruct the volume using box-constrained least squares with isotropic three-dimensional total variation regularisation [21]. The resulting optimisation problem is solved using the alternating direction method of multipliers (ADMM) [4].

Multi-slice plug-and-play. PnP reconstruction combines the physical measurement model with denoisers acting as implicit priors [11, 24]. Following multi-slice fusion [13], we denoise a shared three-dimensional volume along complementary orientations. For $o \in \{xy, yz, zx\}$, P_o rearranges the volume into two-dimensional slices, which $\mathcal{D}$ processes independently, giving $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$.

We combine measurement consistency, denoising and sparsity through PnP-ADMM. Let $\mathcal{J}$ index the active branches, with penalties ρ_j, auxiliary volumes $\mathbf{v}_j$ and scaled dual variables $\mathbf{d}_j$. Each iteration solves

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathsf{A}\mathbf{x}-\mathbf{y}\|_2^2 + \sum_{j\in\mathcal{J}} \tfrac{\rho_j}{2}\|\mathbf{x}-\mathbf{v}_j^k+\mathbf{d}_j^k\|_2^2 \tag{5}$$

using conjugate gradients. The branches then apply denoising or soft thresholding, followed by

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}.$$

Pretrained PnP uses a natural-image DnCNN [29] across the three slice orientations and a sparsity branch. *Learned PnP* replaces this denoiser with a neuron-trained 2D network shared across orientations and adds a volumetric branch with a neuron-trained 3D network [11, 30]. Both variants retain the same measurement-consistency formulation. We initialise from normalised back-projection and clip the final estimate to $[0, 1]$. Iteration counts, penalties and parameter-selection procedures are specified in the supplement.

Conformal depth intervals. To quantify uncertainty in recovered depth, we construct calibrated intervals around the axial intensity centroids of reconstructed volumes. For each lateral column c containing signal, let $\hat{z}_c$ and z_c denote the reconstructed and reference centroids. We report intervals $\hat{z}_c \pm \lambda$, where λ is a common half-width calibrated on held-out volumes separately for each encoder. For each volume, the loss $L(\lambda)$ measures the signal-weighted fraction of columns whose reference depths fall outside their intervals. Since $L(\lambda)$ is bounded by one and non-increasing in λ, conformal risk control [1] selects

$$\hat{\lambda} = \inf\left\{\lambda : \tfrac{n}{n+1}\hat{R}_n(\lambda) + \tfrac{1}{n+1} \le \alpha\right\} \tag{6}$$

where $\hat{R}_n(\lambda)$ is the mean loss over n calibration volumes. Provided the calibration volumes and the new volume are exchangeable, this gives $\mathbb{E}[L(\hat{\lambda})] \le \alpha$, with expectation over both calibration and the new volume. This controls expected signal-weighted miscoverage; it does not guarantee coverage for each column or specimen, or establish neuronal connectivity.

Bounds without the learned prior. A reconstruction selects one volume, even when several volumes are consis-

tent with the measurements. To assess this ambiguity without relying on the learned reconstruction prior, we use set-membership estimation [8, 16]. Given the intensity bounds $\mathbf{x} \in [0,1]^N$, we define the measurement-consistency set

$$\mathcal{C}(\mathbf{y}) = \{\mathbf{x} \in [0,1]^N : \ \ell_i \le (\mathsf{A}\mathbf{x})_i \le h_i\},$$

where the intervals $[\ell_i, h_i]$ jointly contain the noise-free measurements with a stated probability under the assumed noise model. For a non-empty consistency set, minimising and maximising x_j over $\mathcal{C}(\mathbf{y})$ gives the smallest and largest intensities consistent with these constraints. Both bounds are computed by linear programming. Their difference measures the remaining ambiguity in that voxel's intensity.

The connection to optical learning can be understood through a two-voxel example. Let $\mathbf{a}_1, \mathbf{a}_2$ be unit-norm measurement columns with correlation ρ, where $|\rho| < 1$. For coefficient perturbations satisfying $\|\mathbf{a}_1\delta x_1 + \mathbf{a}_2\delta x_2\|_2 \le \varepsilon$, allowing δx_2 to vary freely gives $|\delta x_1| \le \varepsilon/\sqrt{1-\rho^2}$. Highly correlated responses therefore permit greater ambiguity. This provides intuition for reducing depth correlations, rather than a guarantee for the full system, which couples many voxels, includes intensity constraints, and uses raw measurement columns rather than the centred training signatures. We compute the actual bounds on reduced fields where the linear programs are tractable.

5. Experimental setup

All experiments use the simulated fixed-focus fluorescence microscope described in Sec. 3. Excitation and emission wavelengths are set to 532 nm, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has $0.32\,\mu$m features and is positioned $6.6\,\mu$m before the intermediate illumination image plane, in specimen-equivalent coordinates. We use 256 illumination directions and evaluate the optical fields on a $0.08\,\mu$m grid, with $0.32\,\mu$m analysis sampling and $0.64\,\mu$m reconstruction voxels. The model assumes linear fluorescence in a homogeneous, non-scattering medium.

Volumes and compression. We reconstruct volumes on a $16 \times 48 \times 48$ voxel grid ($N = 36{,}864$), covering $30.7 \times 30.7 \times 10.2\,\mu\text{m}^3$ (lateral $\times$ lateral $\times$ axial). Each camera image contains 12×12 samples. Using $k = 256$, 64, or 32 illumination directions gives measurement compression factors $C = N/(144k)$ of 1, 4, and 8, respectively. For $C > 1$, we evaluate five randomly selected direction subsets. Reference volumes are obtained from two-photon stacks of mouse pyramidal neurons in the public DEEP^2 dataset [25], min–max normalised and resampled to the analysis grid.

Dataset splits and validation. The test set contains six neuronal cells from three animals, with two prespecified crops per cell selected by a structure score. With reconstruction parameters fixed, we randomly selected eight additional crops per cell, giving 60 test crops for the cross-decoder evaluation (Sec. 6.2). Unless otherwise stated, other full-field experiments use the 12 prespecified crops. Neuron-trained denoisers are trained on 459 crops from a fourth animal. For conformal calibration, we use 35 non-overlapping crops from a held-out stack of that animal, with 240 overlapping crops used for sensitivity analysis. Reconstruction parameters are selected separately for each encoder and compression setting using two vascular crops from a separate stack acquired from a different animal. No test volumes are used for training, tuning, or calibration.

Encoders and photon budget. We compare three random phase-plate realisations, a defocused binary amplitude mask at the plate plane, angular scanning without a plate, and an in-focus DMD with random binary patterns. All encoders share the same relay, pupil, detection model and forward-operator assembly. Excitation is simulated over a field larger than the specimen to avoid beam-edge effects and normalised to equal total deposited dose. For k illumination directions, photon counts follow $\text{Poisson}(\kappa_0\, 256\, \bar{y}_i/k)$, where $\bar{y}_i$ is the noiseless measurement. The factor $256/k$ increases exposure per image as the number of directions decreases, maintaining a fixed total excitation dose. For each specimen, κ_0 is calibrated using random realisation 0 to give 1000 expected photons per pixel per image at the mean signal for $k = 256$, and is shared across encoders. The learned plate is initialised from this realisation, optimised within the specimen field, and trained with two seeds.

Metrics and statistics. We evaluate reconstruction quality using PSNR, foreground SSIM, and local depth error. Local depth error measures the absolute difference between reconstructed and reference axial intensity centroids at each lateral position, averaged with weights proportional to the reference column brightness. Only columns exceeding 10 % of the maximum reference column brightness are included. To contextualise this metric, we use two depth-blind references: an oracle that preserves the exact lateral structure but assigns each column the crop-averaged axial profile, and an estimate uniform across depth. Comparisons are paired across specimens, illumination-direction subsets, excitation dose, and noise protocol. Since crops from the same cell are dependent, we construct confidence intervals by resampling cells with replacement from the six cells (10,000 bootstrap replicates).

Reduced-field experiments. Computing voxelwise measurement-consistency bounds requires solving two linear programs per voxel, making full-field evaluation computationally expensive. We therefore use a reduced

	random plate		learned plate		depth change vs random		DMD
decoder	PSNR	depth	PSNR	depth	seed 0 [95 % CI]	seed 1	depth†
TV	36.70	0.259	38.29	0.150	−42 % [−46, −38]	−43 %	2.31
pretrained PnP	35.72	0.490	37.56	0.224	−54 % [−57, −51]	−55 %	2.17
learned PnP	38.01	0.197	39.60	0.139	−30 % [−32, −27]	−31 %	2.25

Table 1. **The learned plate under three decoders** ($C = 8$, mean over 60 test crops from six cells × 5 measurement draws, equal dose; PSNR in dB, local depth error in μm). Plates are paired on the same objects, direction subsets, dose and noise protocol; depth change with cell-bootstrap 95 % interval. †DMD on the 12 prespecified crops (flat estimate 2.15 μm).

field with the same optical model, containing $4 \times 12 \times 12$ reconstruction voxels and 3×3 camera samples per image, simulated using 32 illumination directions. For this analysis, we assume Gaussian noise at the 15 dB reference setting and construct measurement intervals with 95 % simultaneous coverage using a Bonferroni correction ($\alpha = 0.05$). We evaluate 24 synthetic phantoms with three noise realisations each, together with 48 windows extracted from the neuronal test volumes. Further details are provided in the supplement.

6. Results

6.1. Depth encoding under angular illumination

We first test the untrained depth-encoding mechanism (Fig. 3), comparing the random phase plate with a defocused amplitude mask, a plate-free scan, and an in-focus DMD under matched conditions. The phase plate produces depth-dependent lateral shear, with measured slopes of 0.74–0.94 times the thin-screen prediction and a pivot near the plate's conjugate plane. Excitation sequences at fixed lateral positions decorrelate within approximately 1 μm of axial separation.

Using 32 illumination directions, maximum-likelihood localisation of point emitters with unknown brightness gives depth misassignment rates of 1.5–1.8 % for the phase plates, compared with 4.4–5.3 % for amplitude masks, 25 % for the DMD, and 89 % without a plate. In neuronal reconstruction with pretrained PnP, the phase plate achieves 0.36 μm local depth error at $C = 1$, while the DMD remains near the flat-depth baseline (2.17 versus 2.15 μm). The amplitude mask is competitive in depth error, although the phase plate achieves 0.4–1.2 dB higher PSNR at $C = 8$. These results establish defocused angular illumination as an effective depth encoder, with the random phase plate providing a strong baseline for learning.

6.2. Decoder-free learning improves depth recovery

Figure 1 and Tab. 1 compare the learned plate with its random initialisation across TV, pretrained PnP, and neuron-trained PnP. At $C = 8$ and equal excitation dose, mean local depth error decreases by 42 %, 54 %, and 30 %, respectively, over 60 neuronal crops, with cell-bootstrap intervals excluding zero. The evaluation includes 48 randomly selected crops added after decoder parameters were fixed, alongside the 12 prespecified crops. Depth error improves in all crop–decoder comparisons except one. PSNR increases by 1.6–1.8 dB, and a second training seed yields comparable gains.

Fewer measurements. On the 12 prespecified crops, the learned plate reduces depth error at all tested compression factors ($C = 1, 4, 8$). At equal total excitation dose, the learned plate with 32 illumination directions outperforms the random plate with 256 directions under pretrained PnP, reducing depth error by 35 % (95 % CI: 28–42 %). TV also achieves a lower mean error (0.153 versus 0.187 μm), while neuron-trained PnP gives comparable means (0.164 versus 0.161 μm), without sufficient evidence to establish equivalence.

6.3. Limits of reconstruction without depth encoding

The in-focus DMD provides a control for examining whether stronger reconstruction priors can compensate for poor axial discrimination. At $C = 8$, all three decoders yield local depth errors of 2.17–2.37 μm, close to the flat-depth reference of 2.15 μm. On the additional 48 randomly selected crops, neuron-trained PnP remains near the flat-depth reference (1.81 versus 1.93 μm). A second, independent DMD pattern set reproduces the results within 0.07 μm in depth error, suggesting that the limited depth recovery is not specific to the initial pattern realisation.

To separate lateral and axial improvements, we decompose each column's reconstruction error into a depth-averaged component and a zero-mean axial component. At $C = 8$, neuron-trained PnP reduces the random plate's lateral and axial errors to 0.48 and 0.71 times those of pretrained PnP, respectively. For the DMD, the corresponding ratios are 0.77 and 1.04, with axial error accounting for 99 % of the total squared error (median over the prespecified crops). Figure 4 illustrates this distinction: the stronger prior sharpens lateral structures in the DMD reconstruction without recovering their depth.

6.4. Comparison of optical learning objectives

We compare our depth-decorrelation objective with worst-case coherence [26], depth Fisher information [22], and an uncentred variant of our objective (Tab. 2). All plates are trained from the same initialisation with matched sampling and optimisation budgets. Although all objectives improve upon the random plate, ours achieves 12–14 % lower depth error than coherence and Fisher information at $C = 8$, and 14–19 % at $C = 1$. The advantage persists across both training seeds and all three decoders, with 12–22 % lower

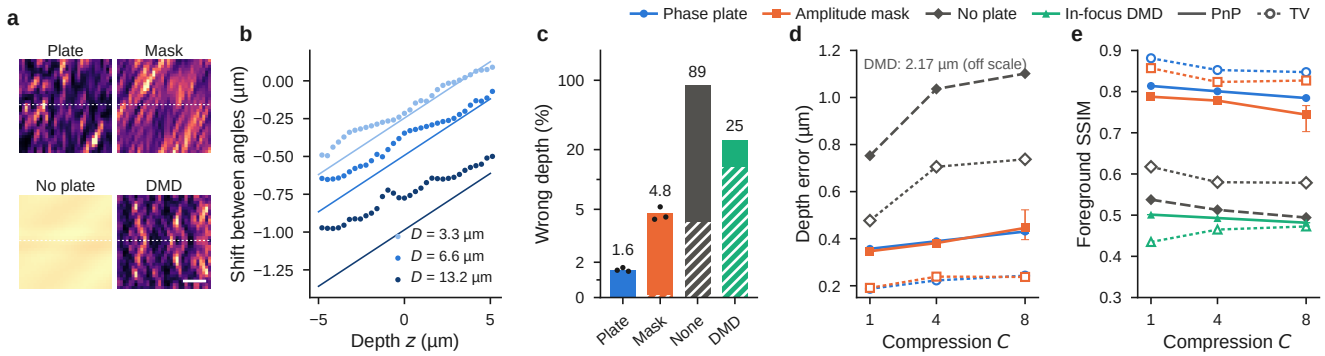


Figure 3. **Defocus under the angle scan encodes depth; in-focus patterns and a plain scan do not** (full-size field, equal deposited excitation dose, overfilled illumination aperture; phase plate: the random plate). (a) One excitation pattern per encoder, xz section through the specimen depth ($\pm 5\,\mu$m, focus dotted), each tile scaled to its own 99.5th percentile. (b) Lateral shift of the plate's excitation between two scan angles at every depth, from the cross-correlation peak, for plate defocus $D = 3.3$, 6.6 and $13.2\,\mu$m; lines: thin-screen prediction of Sec. 3. (c) Point emitters assigned a wrong depth by maximum likelihood with unknown brightness at $k = 32$ (dots: three plate and three mask realisations; hatched: mirror depth). (d, e) Reconstructions of the 12 prespecified neuron crops from 6 cells: local depth error and foreground SSIM versus compression, pretrained PnP (solid, no plate dashed; plate and mask at $C = 4, 8$: mean and range over three realisations) and TV (dashed); the DMD's depth error of $2.17\,\mu$m is off scale. No plate: simulated aperture four times the field in (a, d, e), twice in (c). All values are tabulated in the supplement.

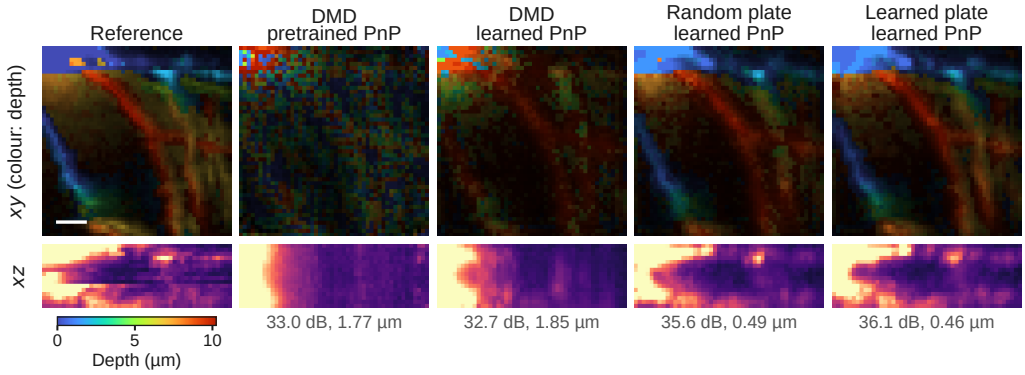


Figure 4. **Same decoder, different encoders** ($C = 8$, prespecified display crop with the largest reference depth spread, draw 0, identical display). Top: lateral maximum-intensity projection, colour coding the depth of the brightest voxel; bottom: xz projection; below: PSNR and local depth error of this crop. Learned PnP sharpens the DMD's lateral structure but not its depth. On this crop the learned plate improves on the random plate only slightly (0.46 against $0.49\,\mu$m); the cohort mean is in Tab. 1. Learned-plate column: reconstruction from the conformal study's rerun, which reproduces the recorded row to 0.02 dB. Bar: $5\,\mu$m.

depth error at $C = 8$. Centring the signatures has negligible effect. These results favour our objective under the tested conditions, rather than establishing its general superiority over alternative criteria.

6.5. Depth uncertainty and measurement ambiguity

Conformal depth intervals. At $C = 8$ and a target miscoverage of $\alpha = 0.1$, the learned plate reduces the calibrated depth-interval half-width from 0.646 to $0.474\,\mu$m (26.7 %), with a second training seed achieving $0.486\,\mu$m (Tab. 3). The observed signal-weighted miscoverage is 0.075 for the learned plate and 0.060 for the random plate, both below the target. Conformal risk control guarantees expected miscoverage only under exchangeability; since calibration and test volumes come from different animals, this assumption is not assured. The observed coverage therefore supports, but does not establish, transfer across animals.

Bounds without the learned prior. In the reduced-field synthetic-phantom experiments, the learned plate reduces mean voxelwise measurement-consistency bound width by 21 % relative to the random plate (Tab. 3). This accompanies a reduction in mean depth correlation from 0.868 to 0.761, consistent with the two-voxel intuition of Sec. 4. However,

	$C=1$		$C=8$		ours vs this, $C=8$
plate objective	PSNR	depth	PSNR	depth	depth [95 % CI]
none (random plate)	39.18	0.161	37.11	0.221	−26 % [−35, −18]
worst-case coherence	39.95	0.135	38.20	0.186	−12 % [−16, −9]
depth Fisher information	40.15	0.128	38.15	0.190	−14 % [−19, −10]
depth decorrelation, uncentred	40.86	0.108	38.85	0.164	0 % [−1, +1]
depth decorrelation (ours)	40.82	0.109	38.83	0.164	–

Table 2. **Plate objectives** with the same start, sampling, step budget and seed, decoded with learned PnP (12 prespecified test crops; depth in μm). Each plate has its own validation tuning. Last column: depth change of our plate against each row, paired by crop, with cell-bootstrap interval. Second seed and other decoders: supplement.

	conformal depth interval ($\alpha = 0.1$, $C = 8$)			bound width
plate	$\pm\hat{\lambda}$ (μm)	vs random [95 % CI]	test miscov.	reduced field
random	0.646	–	0.060	1
learned, seed 0	0.474	−26.7 % [−30.1, −22.9]	0.075	0.79
learned, seed 1	0.486	−24.8 % [−29.6, −21.6]	0.060	0.79

Table 3. **Depth intervals and bounds.** Conformal half-width of learned-PnP reconstructions, calibrated on 35 non-overlapping held-out crops (calibration-bootstrap 95 % interval), and the realised miscoverage on the 12 prespecified test crops × 5 draws (signal-weighted, target $\alpha = 0.1$). Bound width: mean per-voxel width over $\mathcal{C}(\mathbf{y})$ on the reduced field (15 dB, 72 queries), relative to the random plate.

on the 48 neuronal windows, the bounds remain uninformative across the tested noise levels, spanning the full admissible intensity range for every voxel. Thus, optical learning reduces measurement ambiguity without establishing unique recovery of dense neuronal structures.

6.6. Alignment tolerance and calibration

We evaluate the sensitivity of the learned plate to optical alignment errors. At $C = 8$, plate-distance offsets of $\pm 0.5\,\mu$m retain depth-error reductions of 26–28 % after recalibrating the forward model, while illumination-pointing errors up to 0.01 in direction cosine retain reductions of 18–22 % without recalibration. In contrast, uncalibrated plate-distance and lateral-position errors increase depth error in both plates by 0.15–$0.41\,\mu$m and eliminate the learned plate's advantage. These results highlight the importance of accurate optical registration for preserving the benefits of learned illumination.

7. Discussion and limitations

Decoder-free optical design. Our results show that improving depth discrimination at the measurement stage can benefit different reconstruction methods without coupling optical training to a decoder. The learned plate reduces depth error under TV and two PnP decoders, despite being optimised without specimen examples or reconstruction feedback. These gains are not tied to a single decoder, although the two PnP methods share an underlying framework. Unlike end-to-end optical design, which optimises a particular optics–decoder combination, our approach directly targets depth discrimination in the measurements. We do not establish superiority over end-to-end learning or transfer to arbitrary reconstruction methods.

Depth recovery and measurement ambiguity. The learned plate improves depth placement, and conformal calibration turns the improvement into narrower intervals with an expected-risk guarantee under exchangeability. It does not make neuronal topology identifiable: at eight-fold compression no plate determines dense neuronal windows from the measurements alone (Sec. 6.5). In supplementary experiments, conformal connectivity calls on these reconstructions missed their nominal error rate, and query-adaptive direction selection did not meet its prespecified criterion. Connections drawn from such reconstructions remain the prior's choice.

Physical assumptions and implementation. Our results are based entirely on simulations of linear fluorescence in a homogeneous, non-scattering medium with idealised optical components. Scattering, aberrations, and other experimental imperfections may alter the depth-dependent illumination patterns and the depth discrimination they provide. Although the alignment simulations show that the learned gains can survive moderate perturbations when the optical model is updated, uncalibrated misalignment can eliminate those gains. Experimental validation therefore requires accurate characterisation of the realised optical system and its measurement operator.

Data and statistical limitations. Our evaluation uses 60 neuronal crops from six cells across three animals. Crops from the same cell are correlated, and the cell-bootstrap confidence intervals remain approximate with only six cells. The objective comparison and depth-interval evaluation use 12 prespecified crops, while optical training was repeated with two seeds. The denoisers were trained on a fourth animal, with conformal calibration performed on a held-out stack from that animal. Since the test volumes come from different animals, exchangeability is not guaranteed, and empirical interval coverage does not establish a cross-animal guarantee. Broader evaluation across specimens and imaging conditions is needed to assess generalisation.

8. Conclusion

We present a decoder-free approach to learning depth-encoding illumination for fixed-focus three-dimensional fluorescence microscopy. By optimising depth discrimination directly through the optical forward model, we learn a phase plate without specimen examples or a reconstruction solver.

In simulations, the learned plate reduces depth error by 30–54 % across total variation and two plug-and-play reconstruction methods, while narrowing calibrated depth intervals by 27 %. The tested in-focus DMD baseline, by contrast, leaves substantial depth ambiguity despite stronger reconstruction priors. These results demonstrate that improving optical measurements can benefit different reconstruction methods without tailoring the encoder to a particular decoder.

LLM Usage Statement

LLMs were used for grammar and fact checking.

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